Permutations KEY

Name _____

The counting principle. If one choice has m options and a second choice has n options, there are $m \cdot n$ ways to make both choices.

Example: 3 shirts and 4 pairs of pants make $3 \cdot 4 = 12$ outfits.

Arranging. In how many ways can 4 students sit in 4 chairs in a row?

Chair 1	Chair 2	Chair 3	Chair 4
4 choices	3 left	2 left	1 left

$4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways

Factorial: $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$ read "n factorial"

$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$ $1! = 1$ $0! = 1$ (there is exactly one way to arrange nothing)

Permutations. A **permutation** is an arrangement where **order matters**.

Example: A club of 10 students picks a president, a vice president, and a secretary. How many ways?
$10 \cdot 9 \cdot 8 = 720$ ways. (Ana as president and Ben as vice president is different from Ben as president and Ana as vice president.)

Permutation formula: $${}_nP_r = \frac{n!}{(n-r)!}$$ arrange r things chosen from n things

$${}_{10}P_3 = \frac{10!}{7!} = \frac{10 \cdot 9 \cdot 8 \cdot \cancel{7 \cdot 6 \cdots 1}}{\cancel{7 \cdot 6 \cdots 1}} = 720$$

${}_nP_r$ is read "n permute r." Some books write $P(n, r)$.

Repeats allowed? 3-letter codes from 26 letters:
No repeats: $26 \cdot 25 \cdot 24 = 15{,}600$ (a permutation)
Repeats allowed: $26 \cdot 26 \cdot 26 = 26^3 = 17{,}576$ (not a permutation; use the counting principle)

Work with a partner. Compare answers when you finish each problem.

1. Find each value.

   (a) $5!$ $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$

   (b) $\frac{6!}{4!}$ $6 \cdot 5 = 30$

   (c) $0!$ 1

   (d) ${}_8P_3$ $8 \cdot 7 \cdot 6 = 336$

2. You have 5 shirts, 3 pairs of pants, and 2 pairs of shoes. How many outfits can you make?
   $5 \cdot 3 \cdot 2 = 30$

3. In how many ways can 6 different books be arranged on a shelf?
   $6! = 720$

4. 8 runners are in a race. In how many ways can gold, silver, and bronze medals be given?
   ${}_8P_3 = 8 \cdot 7 \cdot 6 = 336$

5. A phone code has 4 digits (0 to 9).

   (a) How many codes are there if digits can repeat? $10 \cdot 10 \cdot 10 \cdot 10 = 10{,}000$

   (b) How many codes are there if no digit repeats? ${}_{10}P_4 = 10 \cdot 9 \cdot 8 \cdot 7 = 5{,}040$

6. In how many ways can you arrange the letters of the word MATH? $4! = 24$

7. A club of 12 members picks a president and a treasurer. One person cannot have both jobs. How many ways?

   ${}_{12}P_2 = 12 \cdot 11 = 132$

8. Explain why ${}_{10}P_3$ is the same as $10 \cdot 9 \cdot 8$.

   $\frac{10!}{7!}$: everything from 7 down to 1 cancels, so only $10 \cdot 9 \cdot 8$ is left. That matches 10 choices, then 9, then 8.